

The empirical recurrence for OEIS A250672, recovered from its tail

Adrian Perez Fontelles

Independent researcher

8 September 2026

Abstract

OEIS A250672 records “Empirical recurrence of order 47” and keeps the recurrence itself in a linked file rather than in the entry, so the conjecture cannot be read off the entry at all. It can still be settled. The entry counts arrays under a condition local to a bounded window of lines, which makes the count a walk count on a finite digraph and forces a monic integer recurrence of order at most the number $S' = 880$ of states with distinct futures. Berlekamp–Massey on enough exact terms therefore returns the minimal such recurrence, and on the tail beginning at $n = 17$, so the recurrence is proved for every $n \geq 64$. Its last coefficient is $c_{47} = 10319560704 \neq 0$, so no further tail satisfies anything shorter; any order-47 recurrence the sequence satisfies from any point on must then have the same characteristic polynomial. The entry’s recurrence is the one displayed here, whatever its linked file contains, and it is proved.

2020 Mathematics Subject Classification. 05A15, 11B37, 15A18.

1 A conjecture that is not written down

OEIS A250672 is “Number of $(n+1) \times (4+1)$ 0..3 arrays with nondecreasing $x(i,j)+x(i,j-1)$ in the i direction and nondecreasing absolute value of $x(i,j)-x(i-1,j)$ in the j direction.”. It has offset 1 and begins

22119, 329226, 4008211, 41503790, 379890972, 3135539502, 23739860017, 167047195040, ...

The entry states:

Empirical recurrence of order 47 (see link above)

As of the “Last modified” line on the live entry (Jul 23 2025 12:40:09, revision 6) this is still recorded as empirical, and it is still open. The coefficients are nowhere in the entry text: what is conjectured is only that a recurrence of that order exists.

2 The count is a walk count

The entry’s condition is local. It constrains a bounded window of consecutive lines of the array and nothing beyond it, so the admissible configurations of one window are the vertices of a finite digraph, an edge joins u to v when v may follow u , and an admissible array is exactly a walk. Writing M for the adjacency matrix and ι, τ for the indicator vectors of the admissible first and last windows,

$$a(n) = \iota^\top M^{j(n)} \tau$$

for a linear reindexing j fixed by the entry’s shape. States with identical futures contribute identically to that product and may be merged without changing any count; after merging $S' = 880$ states remain, and it is that bound every computation below uses.

Lemma 1. *a satisfies a monic linear recurrence with integer coefficients of order at most $S' = 880$.*

Proof. By the Cayley–Hamilton theorem $M^{S'}$ is an integer combination of $I, M, \dots, M^{S'-1}$, and multiplying that relation on the left by $\iota^\top M^j$ and on the right by τ turns it into a relation among $S' + 1$ consecutive values of a . $\square$

3 Why the question has to be asked of a tail

Running Berlekamp–Massey on the sequence from its first term returns an order LARGER than 47: the opening terms do not satisfy the recurrence, and a recurrence that holds only eventually always looks longer when the exceptional terms are included. That is not evidence against the entry. An OEIS empirical recurrence is asserted for n beyond some index, not from the first term, so the question has to be asked of the tails. It was asked of each tail in turn, and the tail beginning at $n = 17$ is the first whose minimal order is exactly the stated 47.

Lemma 2. *A sequence known to satisfy some linear recurrence of order at most S' is determined, together with its minimal recurrence, by any $2S'$ consecutive terms: Berlekamp–Massey applied to them returns that minimal recurrence exactly.*

Applied to $2S' = 1760$ exact terms from $n = 17$ on, it returns order 47 — the order the entry states — with integer coefficients:

$$a(n) = \sum_{i=1}^{47} c_i a(n-i),$$

c_1	85	c_{13}	220078099986587	c_{25}	10534301265265347583	c_{37}	882658705504970240
c_2	−3520	c_{14}	−912120102137414	c_{26}	−14416650513148669396	c_{38}	−356642285805378560
c_3	94646	c_{15}	3409181245760816	c_{27}	18055345033385931862	c_{39}	125839334647296000
c_4	−1857947	c_{16}	−11532042763674971	c_{28}	−20675010917520802609	c_{40}	−38321868053643264
c_5	28388195	c_{17}	35407561612821767	c_{29}	21620092330424237233	c_{41}	9921303900241920
c_6	−351485258	c_{18}	−98917486552023192	c_{30}	−20613886993891990138	c_{42}	−2140423919861760
c_7	3625160516	c_{19}	251940472893447594	c_{31}	17885472554319266872	c_{43}	374339484647424
c_8	−31775500793	c_{20}	−585949133097272841	c_{32}	−14087496956805608704	c_{44}	−50978486550528
c_9	240308137061	c_{21}	1245929752516059601	c_{33}	10043687410374408640	c_{45}	5070774140928
c_{10}	−1586583733268	c_{22}	−2424325164696168502	c_{34}	−6458803520111236480	c_{46}	−327646052352
c_{11}	9230776455434	c_{23}	4319265571108099732	c_{35}	3730564819003018240	c_{47}	10319560704
c_{12}	−47684749442603	c_{24}	−7048233124870863019	c_{36}	−1925521108683522560		

4 The recurrence, and the identification

Theorem 3. *The recurrence above holds for every $n \geq 64$, and it is the recurrence OEIS A250672 records.*

Proof. That it holds is Lemma 2 applied to the tail from $n = 17$, whose minimal recurrence it is.

For the identification, write $q_{\min}$ for the characteristic polynomial of that minimal recurrence, $x^{47} - c_1 x^{46} - \dots - c_{47}$. Its constant term is $-c_{47} = -10319560704$, which is not zero, so $q_{\min}(0) \neq 0$ and $q_{\min}$ has no root at the origin. A solution of a constant-coefficient recurrence is a combination of terms $n^k \lambda^n$ with λ a root of its characteristic polynomial, and only the components with $\lambda = 0$ vanish after finitely many terms. Since $q_{\min}$ has none, no component of a dies out, and the minimal recurrence of every further tail is again $q_{\min}$ — the order cannot drop below 47 at any later starting point.

Now let q be the characteristic polynomial of whatever recurrence the entry asserts. The entry asserts it has order 47, so $\deg q = 47$, and it is monic. It holds from some index t on. From $\max(t, 17)$ on, the sequence satisfies both q and $q_{\min}$, and the minimal polynomial of that tail is $q_{\min}$ by the previous paragraph; therefore $q_{\min} \mid q$. Both are monic of degree 47, so $q = q_{\min}$. $\square$

5 Verification

Four checks, all in exact integer or rational arithmetic.

First, the walk model reproduces all 16 terms the entry publishes, exactly. This is what ties the model to the entry, and it is the only place where reading the entry’s English enters the argument.

Second, the order was computed twice by independent routes: once modulo a 61-bit prime, where every intermediate is a machine word, and once over $\mathbb{Q}$ in exact rational arithmetic. Both return 47, and the exact run additionally confirms every coefficient is an integer — had any been a proper fraction the recurrence would not be the entry’s.

Third, the recovered recurrence was evaluated directly on the entry’s own published terms, with no matrices and no Berlekamp–Massey involved. It holds at each of the 0 indices where the entry publishes enough earlier terms to test it.

Fourth, the last coefficient was checked to be nonzero, which is what the identification above turns on; it is 10319560704.

Remark 4. The conjecture’s own statement was never read. The entry pins it down to the family of order-47 recurrences this sequence satisfies from some point on, and that family turns out to have exactly one member.

References

- [1] The OEIS Foundation, *The On-Line Encyclopedia of Integer Sequences*, <https://oeis.org>, 2026. Sequence A250672.
- [2] J. L. Massey, Shift-register synthesis and BCH decoding, *IEEE Trans. Inform. Theory* **15** (1969), 122–127.
- [3] R. P. Stanley, *Enumerative Combinatorics*, Volume 1, 2nd ed., Cambridge University Press, 2011. (Transfer-matrix method, Section 4.7.)
- [4] M. Kauers and P. Paule, *The Concrete Tetrahedron*, Springer, 2011. (C-finite sequences and their closure properties.)
- [5] R. A. Horn and C. R. Johnson, *Matrix Analysis*, 2nd ed., Cambridge University Press, 2013.